Experiment No. 4

Title:

File Management and Input/Output Redirection in Linux

Aim:

To study and practice essential Linux file management commands such as creating, copying, moving, renaming, viewing, and deleting files, and to understand the use of input and output redirection for managing command outputs efficiently.

Theory:

File Management Commands

Linux offers several commands to manage files:

- touch – Create empty files
- cp – Copy files or directories
- mv – Move or rename files
- rm – Delete files or directories
- cat – Display file contents

Input/Output Redirection

Redirection allows command outputs to be redirected to files instead of the terminal:

- **— Redirect output (overwrite)**
- **Redirect output (append)**
- **< — Redirect input**

These features are essential for scripting and automation.

Commands Used

Commands	Description
touch	touch
cp	cp
mv	mv
cat	cat
>	Output redirection
>>	Append output

Procedure / Steps

Step 1: Login

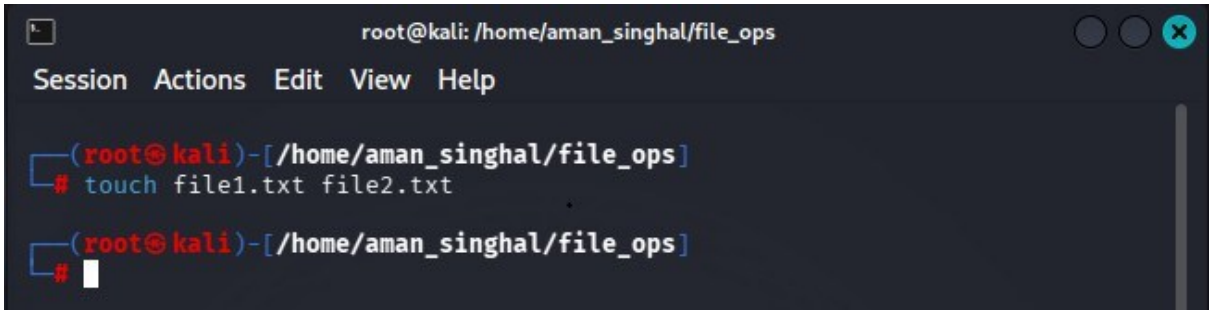
- Log in to the Linux virtual machine and open the **Terminal**.

Step 2: Create a working directory:

- mkdir file_ops
- cd file_ops

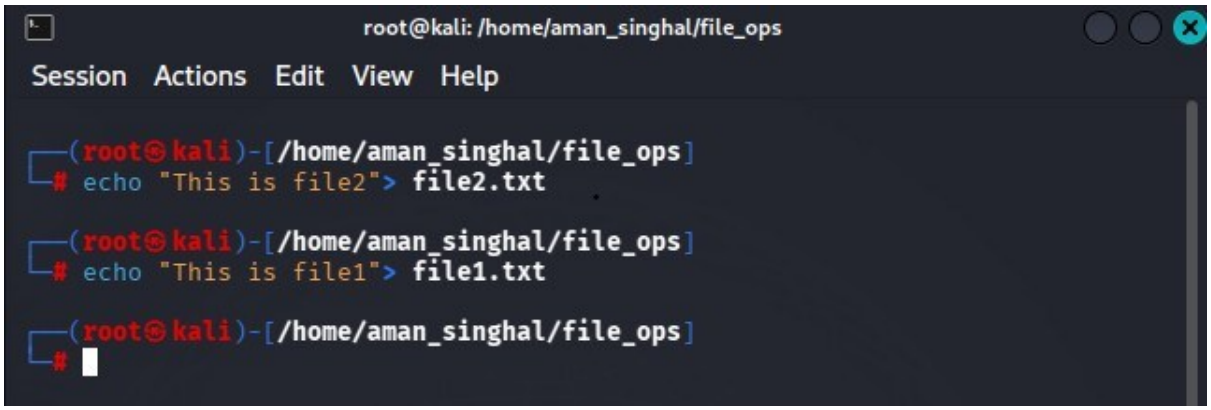
Step 3: Create files using touch:

- `touch file1.txt file2.txt`



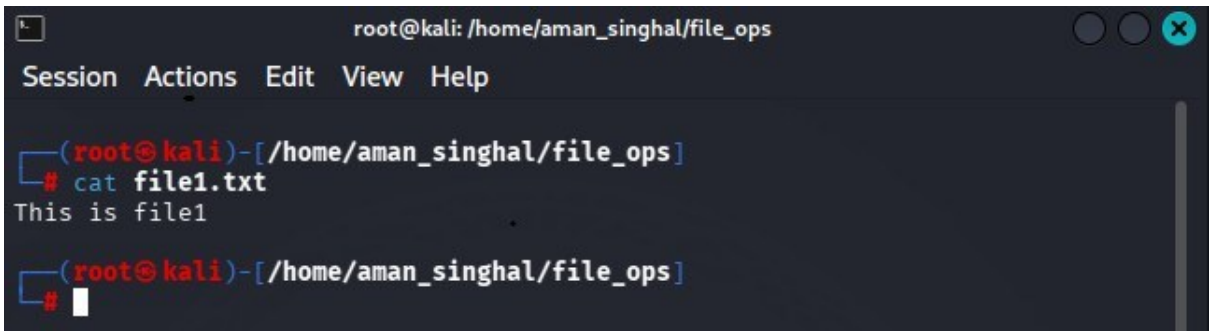
Step 4: Add content to files using redirection:

- `echo "This is file1" > file1.txt`
- `echo "This is file2" > file2.txt`



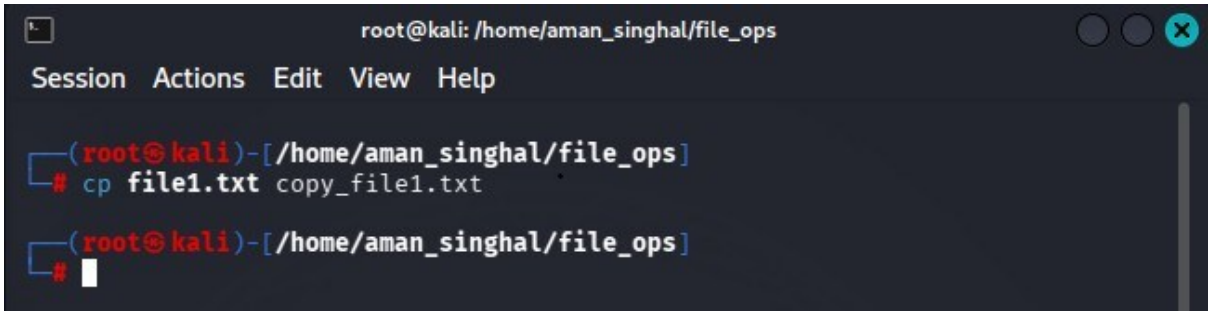
Step 5: Display file contents:

- cat file1.txt



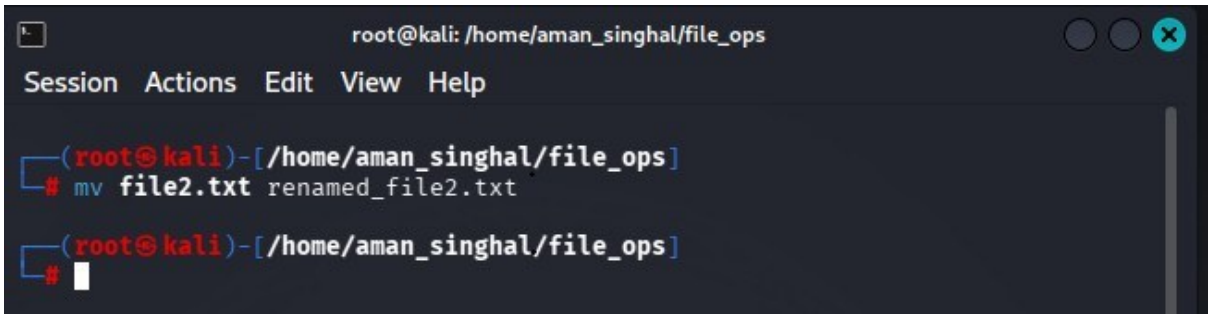
Step 5: Copy a file:

- CP file1 txt copy _ file1.txt



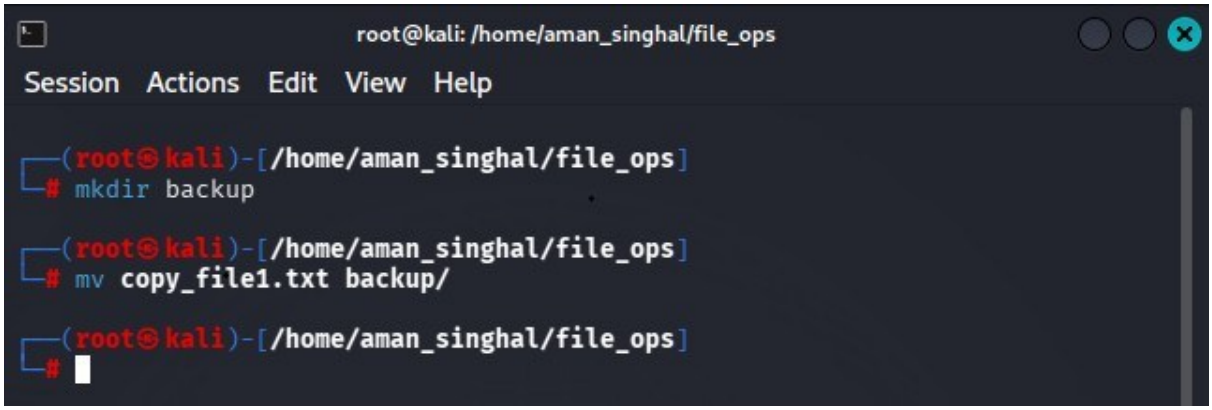
Step 6: Rename a file:

- mv file2. txt renamed file2. Txt



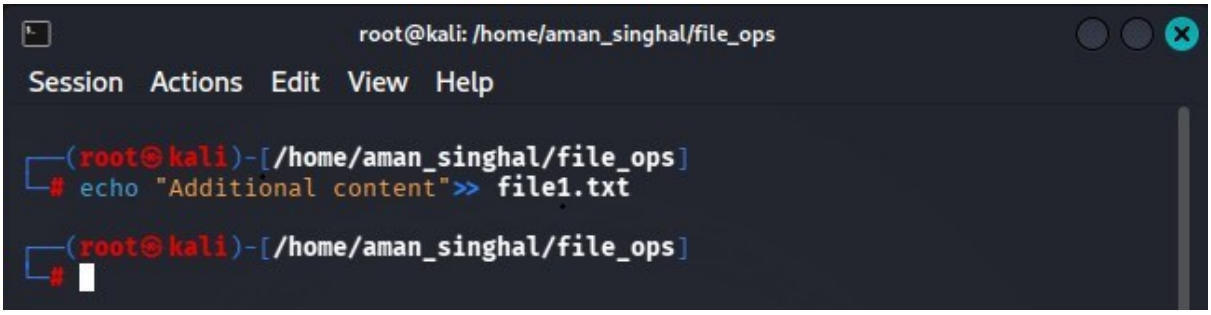
Step 7: Move a file to another directory:

- mkdir backup
- mv copy _ file1.txt backup/



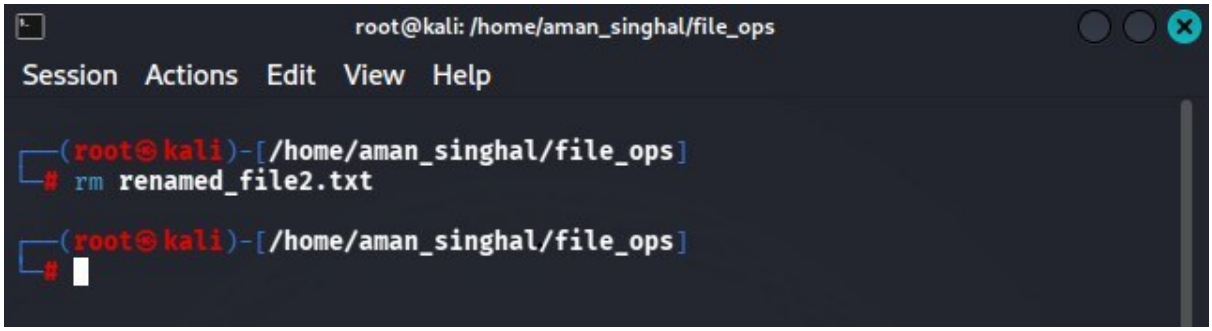
Step 8: Append content to a file:

- echo "Additional content” >> file1.txt



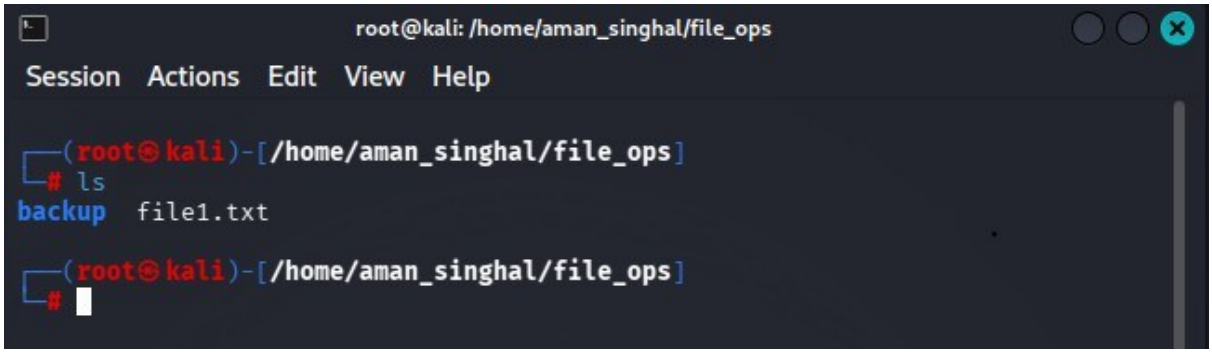
Step 9: Delete a file:

- rm renamed_ file 2. txt



Step 10: Verify changes using:

- ls



Outcomes:

- Create, delete, copy, move, and rename files using Linux commands such as touch, rm, cp, and mv.
- Display and examine file contents using commands like cat, less, more, and head/tail.
- Apply input (<) and output (>, >>) redirection to manage command outputs efficiently.
- Use file management commands safely and effectively in real-world scenarios.
- Automate file operations using basic shell scripting techniques.
- Understand the importance of file management in system administration and cybersecurity tasks.

Result:

- Files are successfully created, copied, moved, renamed, and deleted as instructed.
- File contents are displayed correctly using appropriate commands.
- Output redirection (>, >>) creates or appends to files as expected.
- Input redirection (<) correctly supplies data to commands.
- No unintended data loss occurs during file operations when proper options are used.